

Transplanting Following Non-Native Plant Control in Rocky Mountain Foothills Fescue Grassland Restoration

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ABSTRACT

Disturbed areas within national parks, resulting from historic and current land use activities, can harbor large and diverse populations of non-native plant species. These species must be controlled to prevent their spread and to restore native grassland. Research on alternative revegetation methods is urgently needed since seed based grassland restoration is often unsuccessful. The effectiveness of transplanting to restore foothills fescue grassland following implementation of non-native plant management was investigated at three disturbed sites in Waterton Lakes National Park, Alberta, Canada. Non-native plant abundance was reduced by cutting, glyphosate application, and steaming. Greenhouse grown seedlings of native fescue grassland species (13 forbs, five grasses, two shrubs) were outplanted into plots following management treatments, as three different sizes of container stock. The effects of prior non-native plant management and container size on transplant survival and growth were assessed over two growing seasons. Rhizomatous forbs and bunchgrasses with well-developed root systems had highest survival (> 50% over 14 months). Transplants with 10- to 15-cm rooting depth in cones and root trainers had significantly higher within-year survival than tray transplants with a shallow rooting depth. Transplant survival was improved by glyphosate application to control non-native plants prior to planting. Transplanting was effective for increasing native cover and species richness although high winter mortality reduced this effect. Key species for fescue grassland function were introduced at all sites and persisted for two growing seasons.

Keywords: container size, glyphosate, revegetation, species richness, survival, transplanting

Restoration Recap

- Broadcast seeding is frequently conducted to restore native grasslands, however, when non-native plant species dominate a site, germination and establishment may be low, even with management to control non-native plants.
- The purpose of this study was to determine whether transplanting native forbs, grasses, and shrubs following four treatments to control non-native plants (cutting, glyphosate application, steam, no management) is an effective revegetation method for fescue grassland.
- Transplant survival was highly dependent on species and container type, with high winter mortality for all transplants. Rhizomatous forbs and bunchgrasses with well-developed roots systems and those grown in deeper containers had better performance. Application of glyphosate prior to transplanting significantly improved transplant survival over two growing seasons.
- Transplanting introduced key species that were missing from the sites and enhanced native cover and richness. Results suggest that transplanting native forbs, grasses, and shrubs as cone or root trainer material can be used as part of a program to restore fescue grasslands. Repeated non-native plant management with glyphosate is likely required to improve transplant survival.

 **Supplementary materials are freely available online at:**
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Native grasslands are globally endangered ecosystems that are in decline mainly due to habitat conversion by cultivation, over grazing, and non-native plant species (Gibson 2009). In Alberta, Canada, foothills fescue grasslands once occupied 1.6 million ha, but have been reduced to 17% of their former range (Adams et al. 2009). Ecological restoration provides a solution to this issue but is challenged by lack of revegetation success (Jordan et al.

1988, Berger 1993, D'Antonio and Meyerson 2002). Seeding is the most common revegetation technique used in grasslands (Hedberg and Kotowski 2010). Attempts to establish native species with broadcast seeding have been limited due to lack of available seed and poor seedling establishment, often from competition with non-native plants (Revel 1993, Desserud et al. 2010). Transplanting may improve revegetation with native species because plants have passed their most vulnerable growth stage (Huddleston and Young 2004, Middleton et al. 2010). Transplanting may be particularly beneficial for rare species (Morgan 1999, Guerant and Kaye 2007) and late successional species such as fescue grasses. Transplanting is a promising alternative to seeding, but little information is available on the effective use of transplants in grassland restoration.

Successful transplanting is highly dependent upon non-native species cover (Brown and Rice 2000, Bakker et al. 2003), thus native propagules must be introduced after non-native plant control to increase revegetation success and prevent recolonization by non-native species (Tyser et al. 1998, Bakker et al. 2003, Huddleston and Young 2004, Reid et al. 2009). Herbicides, such as glyphosate, which reduce non-native plant competition, can increase native plant establishment by creating gaps for seedlings and transplants (Wilson and Gerry 1995, Stromberg and Kephart 1996, Ewing 2002, Huddleston and Young 2005, Page and Bork 2005, Jusaitis and Polomka 2008, Holl et al. 2014). Burning and herbicide application can lead to major changes in vegetation and dominance of non-native plants that may outcompete planted species (Lulow et al. 2007). It is unclear how transplanting is affected by more sustainable, alternative, non-native plant control methods such as steaming (application of pressurized water at high temperatures) and cutting, which are desirable to avoid issues such as herbicide resistance and chemical pollution.

Containers for transplants differ in size and shape and can affect survival of planted seedlings. Native perennial grasses and sedge transplants grown in large containers have greater survival than small transplants (Steed and DeWald 2003, Page and Bork 2005). Containers for forbs in grassland revegetation have not been thoroughly investigated, but trends appear similar (Davies et al. 1999, Halsey et al. 2017). Transplants in small containers are known to have higher survival in some circumstances (Bull et al. 2004). Plant size may affect survival below a threshold and for a short time, or perhaps size may not affect survival in native populations adapted to competition with non-native species (Davies et al. 1999, Fehmi et al. 2004, Thetford et al. 2005). Plants in small containers require fewer resources to propagate and can be produced in greater quantities, so determining minimum size for successful native plant establishment will be economical. Size may be less important than propagule type (e.g., seed, root stock, seedling; Bull et al. 2004, Burkhart 2006). The

objectives of this research were to determine effectiveness of transplanting native fescue grassland species grown in different container types following non-native plant control to accelerate restoration.

Methods

Research Sites and Experimental Design

Three study sites were established in disturbed foothills fescue grasslands of Waterton Lakes National Park in the Rocky Mountains of southern Alberta, Canada ([Supplementary Materials](#)). These grasslands occur on Orthic Black Chernozem soils (Coen and Holland 1976) and are dominated by keystone species *Festuca campestris* (foothills rough fescue), *Danthonia parryi* (Parry's oatgrass), *Festuca idahoensis* (Idaho fescue), and forb species. Growing seasons are characterized by cool, wet springs (May to June) and hot, dry summers (July to September). Mean annual precipitation is 808 mm, and annual mean minimum and maximum temperatures are -1.3 and 10.6°C , respectively (Environment Canada 2013). Site 1 was part of a former landfill, and Sites 2 and 3 were gravel pits. Site 1 soil was a sandy loam texture, pH 7.5, organic matter 6.1%, total carbon 3.72%, and total nitrogen 0.24%; Site 2 soil was a sandy loam texture, pH 5.9, organic matter 6.8%, total carbon 4.17%, and total nitrogen 0.29%; and Site 3 soil was a sand to sandy loam texture, pH 7.9, organic matter 4.5%, total carbon 2.87%, and total nitrogen 0.21% (Stover 2013). Vegetation at the three study sites included many non-native species such as the grasses *Bromus inermis* (smooth brome), *Elymus repens* (quack grass), *Poa pratensis* (Kentucky bluegrass), *Poa compressa* (Canada bluegrass), and *Agropyron cristatum* (crested wheatgrass) and the forbs *Centaurea stoebe* (spotted knapweed) and *Cirsium arvense* (Canada thistle). Site 1 had highest non-native vegetation cover at approximately 30%, and Site 3 had the lowest at 7% (Stover et al. 2017).

Treatments were implemented using a randomized block design. Each site was considered a block. Within each block, twelve plots were established in an area of uniform vegetation, topography, and soils. Individual plots were approximately 216 m^2 at Site 1 and 250 m^2 at Sites 2 and 3 and had a 1-m buffer zone surrounding them. Plots were smaller at Site 1 as there was less available area with uniform topography, vegetation, and soils for replicates. Four non-native management treatments (cutting, glyphosate, steam, and no management) were applied to three random plots in each block, once in July 2010 and once in June 2011. Non-native management treatments were applied over the entire plot as either most vegetation was non-native or native species were interspersed closely with non-native species, negating selective application. Herbicides with the active ingredient (ai) glyphosate were foliarly

applied in solution with water according to manufacturer recommendations. In 2010, Roundup Transorb (540 g ai/L [Monsanto Co., St. Louis, MO]) was applied at 18 to 20 L herbicide per hectare. In 2011, Roundup Concentrate Grass and Weed Control (143 g ai/L [Monsanto Co., St. Louis, MO]) was applied at 40 to 46 L herbicide per hectare. Cutting was conducted with grass trimmers, removing as much above ground biomass as possible without disturbing the soil surface; all vegetation was cut except for shrubs, and biomass was left on the plots. Steaming, to kill above ground biomass, was implemented using steam cleaning equipment mounted on a truck. Steam was applied at a temperature of 325°C with a 50-cm-long bar dragged across plots in 2010; steam was released from holes in the bar and sprayed onto vegetation at close proximity. In 2011, steam was applied at 225°C by hose approximately 50 cm from the ground. Duration of exposure for both applications was approximately 15 seconds. Steam was not applied at Site 2 due to limited vehicle access.

Transplanting

Between 29 May and 5 June 2012 (during a period of maximum precipitation in spring), 13 forb, five grass, and two shrub species were transplanted in each plot (Table 1). The same number of plants and species were planted in all plots, including controls. Plants were grown from January to May 2012 from seed that was wild-collected in Waterton Lakes National Park. A diversity of forb and grass species native to the local study area was selected to provide a diverse restored community and because little information was available on individual species. We selected native grasses that were important keystone species to the restoration: *F. campestris*, *D. parryi*, and *F. idahoensis*. *Erigeron* spp. (fleabane) and *Potentilla* spp. (cinquefoil) were identified only to genus as material was a mixture of species.

The three types of transplants were cone seedlings, root trainer seedlings, and tray seedlings (Supplementary Materials). Cone seedlings were grown at Glacier National Park Native Plant Materials Propagation Center in potting soil (peat moss with perlite) in 15.24-cm-deep and 3.80-cm-diameter cones. Root trainer and tray seedlings were grown at a University of Alberta Plant Growth Facility in potting soil. Growing conditions were similar at both propagation facilities. Seedlings at the University of Alberta were started in 30-cm-wide × 50-cm-long × 5-cm-deep trays. A portion of seedlings of four forb and three grass species (Table 1) were transferred into 11.4-cm-deep and 2.5-cm-wide root trainers (Beaver Plastics Ltd., Acheson, AB) after shoots reached approximately 4 cm. The remaining seedlings of six forb and four grass species were transplanted into the field from trays after 4 months. Trays were divided into 5-cm-wide × 5-cm-long × 5-cm-deep sections for transplanting, with number of

seedlings per section dependent on germination. Approximate densities per tray section for forbs were *Achillea millefolium* (common yarrow) and *Anemone multifida* (cutleaf anemone [1]), *Linum lewisii* (wild blue flax [4]), *Penstemon confertus* (yellow penstemon [13]), *Monarda fistulosa* (wild bergamot) and *Potentilla* spp. (cinquefoil [35]), grasses (30) except for *Koeleria macrantha* (june grass [54]). Table 1 provides the number of tray sections planted per species.

A total of 159 transplants (71 forbs, 84 grasses, and four shrubs) were planted in each plot, consisting of 119 cones, 14 root trainers, and 26 trays. Each plot was gridded (1 × 1 m) and transplants randomly assigned to grid squares; some grid squares were not selected and were left empty. Grid squares contained a maximum of one planting in the center of each square, so the minimum distance between transplants was 1 m. At planting, tray seedlings were approximately 5 cm tall, root trainer seedlings 10 cm, and cone seedlings 15 cm. Holes equal to transplant root depth were dug using metal trowels and post hole diggers to minimize impact to surrounding vegetation; a single plant was placed in each hole and soil firmly placed around it. Plantskydd® (Tree World Plant Care Products, Inc., St. Joseph, MO), an organic herbivore deterrent, was applied to all transplants in all plots.

Vegetation Assessments

Transplants were assessed in late June (20 days following planting), late July 2012 (50 days), and late July 2013 (426 days). For assessment, grids were reestablished and plants were located using maps. Each individual transplant was located and it was noted if the plant was in flower. Health was assessed with a 5-point scale; 0 for plants that could not be located, 1 for dead plants (0% live green tissue), 2 for plants with < 25% live tissue, 3 for plants with 25–50% live tissue, 4 for plants with 51–75% live tissue, and 5 for plants with > 75% live tissue.

To determine if transplanting increased native plant cover and species richness, in addition to monitoring individual plants (described above), surveys with quadrats were conducted in each plot prior to planting in 2011 and in July 2012 and 2013 when plants were at peak production. Ten 0.1 m² quadrats were systematically located in each plot. In each quadrat, an ocular estimate of percent canopy cover by species was recorded. Nomenclature followed Kuijt (1982), Lesica (2002), and Moss (1983). Species richness was counted for each quadrat. Species not located within quadrats, but within plots, were recorded during a time bounded survey and used to calculate plot richness. Non-native plant cover was assessed and results are provided in Stover et al. (2017) as it was not the direct focus of this paper. Glyphosate reduced non-native cover the most while cutting had no effect.

Table 1. Transplant survival by container type in the first two years following planting in disturbed foothills fescue grasslands of Waterton Lakes National Park in the Rocky Mountains of southern Alberta, Canada. Sites and non-native management treatments are pooled.

Species	Container Type	Total Number Planted	Survival (%)		
			June 2012	July 2012	July 2013
Forbs					
<i>Achillea millefolium</i>	Root Trainer	33	86	76	71
	Tray	66	33	37	35
<i>Anemone multifida</i>	Tray	66	11	7	1
<i>Artemisia michauxiana</i>	Cone	264	85	87	58
<i>Erigeron</i> spp.	Cone	99	74	67	26
<i>Gaillardia aristata</i>	Cone	462	80	78	38
<i>Galium boreale</i>	Cone	231	57	49	15
<i>Geranium viscosissimum</i>	Cone	66	77	60	26
<i>Heterotheca villosa</i>	Cone	264	87	85	57
<i>Linum lewisii</i>	Root Trainer	33	43	50	20
	Tray	33	23	30	10
<i>Lupinus sericeus</i>	Cone	33	21	11	3
<i>Monarda fistulosa</i>	Cone	132	77	75	34
	Root Trainer	33	61	78	22
	Tray	99	18	21	9
<i>Penstemon confertus</i>	Cone	99	93	76	5
	Tray	33	29	31	22
<i>Potentilla</i> spp.	Cone	165	87	88	46
	Root Trainer	33	79	74	11
	Tray	99	48	43	14
Grasses					
<i>Danthonia parryii</i>	Cone	363	85	76	19
	Root Trainer	132	83	68	11
	Tray	66	38	23	3
<i>Elymus trachycaulus</i>	Cone	429	83	78	41
	Tray	66	21	18	21
<i>Festuca campestris</i>	Cone	495	82	72	19
<i>Festuca idahoensis</i>	Cone	363	85	81	34
	Root Trainer	99	92	85	46
	Tray	99	42	29	21
<i>Koeleria macrantha</i>	Cone	330	88	86	51
	Root Trainer	99	85	84	42
	Tray	231	47	37	34
Shrubs					
<i>Amelanchier alnifolia</i>	Cone	66	95	92	37
<i>Rosa woodsii</i>	Cone	66	84	83	40

Data Analyses

A binary response variable was created with 0 for missing or dead transplants and 1 for alive transplants. A binary response was chosen as there were less than 15% of transplants with a health score of 2 or 3 in 2012 and less than 3% in 2013. Therefore, these scores were combined with other live categories. Survival was calculated as percent of

transplants per plot alive at each assessment date. Winter mortality was calculated as percent of transplants alive in July 2012 but dead by July 2013.

All tests were conducted using SPSS 23 (IBM Corporation, Armonk, NY) and an alpha of 0.05. Chi square tests were conducted to determine effect of non-native plant management treatments, container type and site on transplant survival within years and over winter and

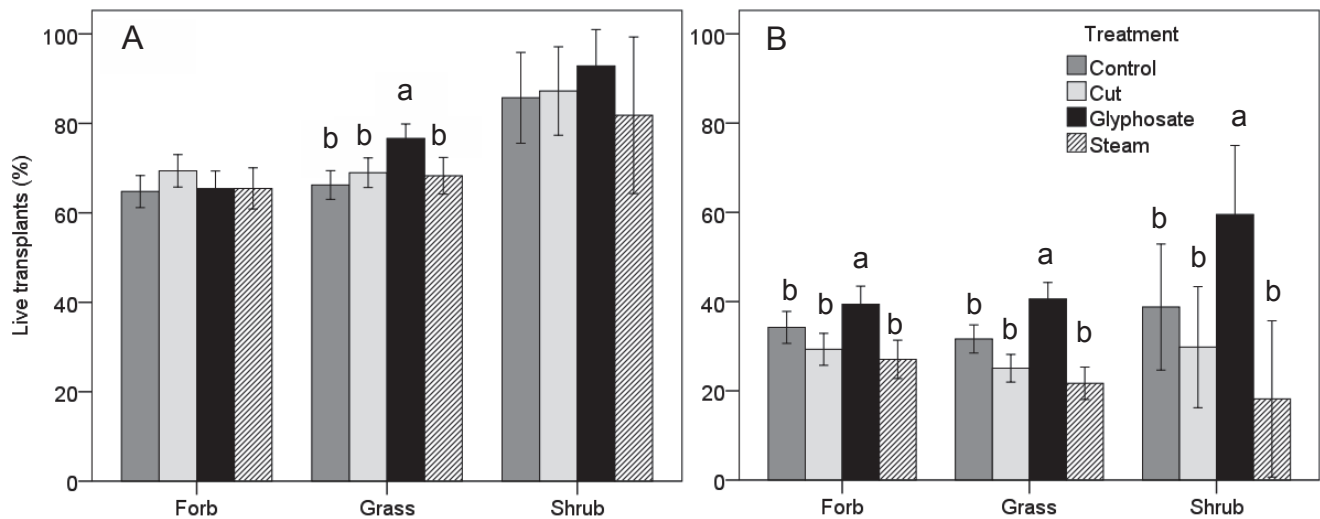


Figure 1. Mean transplant survival (percentage live plants of total planted) at three disturbed sites in (A) July 2012 and (B) July 2013, in disturbed foothills fescue grasslands of Waterton Lakes National Park in the Rocky Mountains of southern Alberta, Canada. Transplants were planted one year following treatments to manage non-native plants. Error bars are 95% confidence intervals. Within functional groups, columns that do not share a common letter are significantly different.

flowering. Analyses were conducted for forb, grass, and shrub species separately. To assess the relationship between container type and transplant survival and flowering, chi square tests were conducted for five forbs and three grasses as others were not planted in different container types (Table 1). To assess the relationship between site and non-native management treatment and transplant survival and flowering, only cones were included in chi square tests to standardize material. Site was included in the analysis to account for any differences due to site conditions. Following a significant chi square test, post hoc comparisons were conducted using residuals (Beasley and Schumacker 1995, Sharpe 2015). Paired t-tests were performed to determine significant changes in native cover and species richness between years.

Results

Transplant Survival and Health

Survival in the first 30 days was greater than 75% for most species, except *Lupinus sericeus* (silky lupine [cone]), *Galium boreale* (northern bedstraw [cone]), *L. lewisii* (root trainer), and all tray transplants (Table 1). Within 60 days there was little change in survival for these species and container types. Of the forbs, *Heterotheca villosa* (hairy false goldenaster [cone]), *Artemisia michauxiana* (Michaux's wormwood [cone]), and *Potentilla* spp. (cone) had best performance in the first growing season, and of the grasses, *F. idahoensis* and *K. macrantha* (cone and root trainer) performed the best: all maintained greater than 85% survival. Shrubs had very high survival in the first season.

Overwinter forb mortality was as high as 94% for *P. confertus* (cone) and as low as 7% for *A. millefolium* (root trainer; data not shown). Grass overwinter mortality was as high as 90% for *D. parryi* (tray) and as low as 38% for *K. macrantha* (tray). Second-year survival was lower than in the first year for forbs, grasses, and shrubs and highly variable (Table 1). Forb survival was 1 to 71%, with *A. millefolium* (root trainer) highest; grass survival was 3 to 51%, with *K. macrantha* (cone) highest. Shrubs had approximately 40% survival in the second season.

Health declined with time during the first growing season (data not shown). *Artemisia michauxiana* and *H. villosa* had considerably more seedlings scoring 4 or greater at each assessment. Health was similar among grasses in the first year. By the second year *Elymus trachycaulus* (slender wheatgrass) and *K. macrantha* cones had considerably more seedlings with a health score of 4 or more. *Elymus trachycaulus*, *Gaillardia aristata* (blanket flower), *P. confertus*, *H. villosa*, and *Erigeron* spp. flowered and produced seed in the first year. Some species such as *M. fistulosa*, *A. michauxiana*, and *G. boreale* recovered after shoot death in spring 2012 by producing new shoots later in summer, and therefore health scores increased.

Effect of Non-Native Management on Transplants

In the first growing season, non-native plant management only significantly affected grass transplant survival (Figure 1). Grass transplants in glyphosate treatments had more live plants than expected relative to other treatments in June (82%) and July (76%) 2012. When partitioned by site, percentage of live forbs was also greatest with glyphosate treatment in July 2012, except at Site 3. In the second year

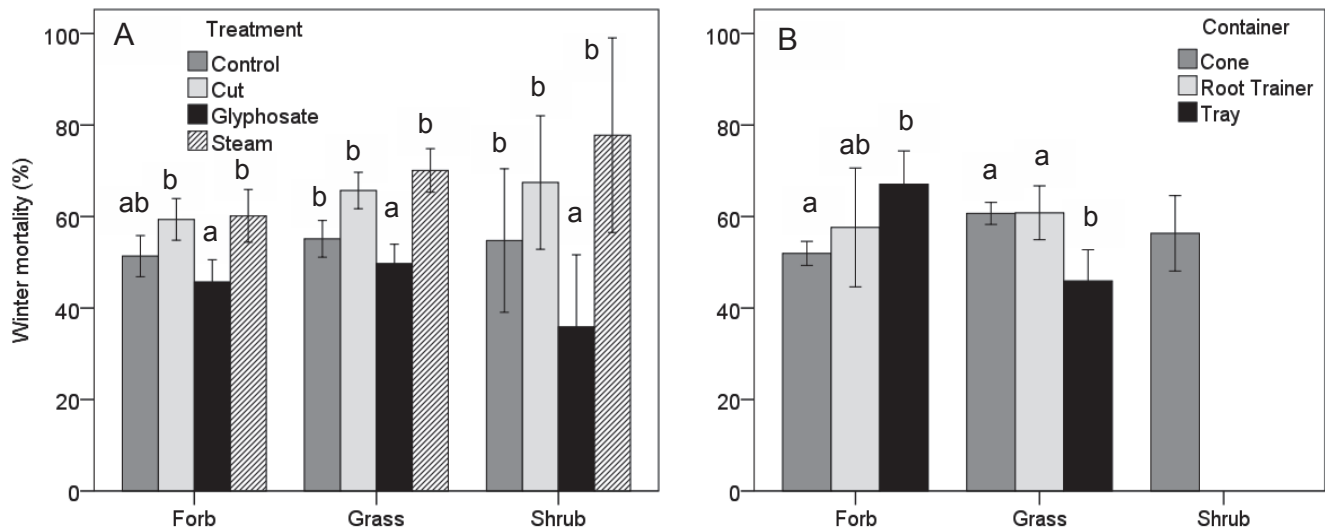


Figure 2. Effect of (A) prior non-native plant management and (B) container type on first winter (2012–2013) mortality. Within functional groups, columns that do not share a common letter are significantly different. Error bars are 95% confidence intervals.

(2013), forb, grass, and shrub survival were significantly greatest in glyphosate treatments. Overall, shrub transplants had higher survival than forbs or grasses at all sites in most treatments in both years.

Overwinter mortality was significantly lowest in glyphosate treatments and highest in steam and cut treatments (Figure 2), except at Site 3, where forb winter mortality in glyphosate was similar to control and cut. Treatment had no effect on forb or shrub flowering, but grass transplant flowering was significantly lower in treatments where vegetation had been cut prior to planting.

Effect of Container Type on Transplants

Container type significantly affected survival of forbs in both years (Figure 3). In the first growing season, forb cone seedlings had significantly higher survival and tray seedlings significantly lower survival than other container types. In the second season there was no difference between forb cone and root trainer seedlings. When accounting for management treatments, forb cone seedlings had significantly greater survival than root trainers within cut treatments only. For grass species, there was no difference in survival between cones and root trainers in both years, however grass tray seedlings had significantly lower survival than both. Treatment had no effect on survival results for grasses.

There was a significant difference in winter mortality among container types for forb and grass transplants (Figure 2). For forbs, cones had lower mortality than trays or root trainers. For grasses, tray seedlings had significantly lower mortality than did cones or root trainers. When partitioned by treatment, these effects were most significant in cut and steam treatments and least in glyphosate and control. Steam treatments had considerably lower grass

mortality for tray transplants than did cones or root trainers (30, 71 and 74%, respectively) relative to control and cut treatments where the difference was 10 to 20%.

Grass flowering was not affected by container type. Flowering frequency was significantly highest in forb cones and significantly lowest in forb tray seedlings (data not shown). Seven of the thirteen cone transplant species flowered, with *A. michauxiana* and *H. villosa* having highest flowering, 18% of seedlings for both. Of the eight forb species planted as tray seedlings only three species (*A. millefolium*, *P. confertus*, *Potentilla* spp.) produced flowers, 2% of forb tray seedlings planted; of the five forb species planted as root trainer seedlings, only one produced flowers (5% of *A. millefolium* seedlings).

Effect of Transplanting on Native Abundance and Diversity

Overall transplanting increased total native cover at each site and in all non-native management treatments relative to the year prior to planting. This was mainly due to an increase in native forb cover at all sites. Significant differences were only found in some sites and treatments (Table 2). Native shrub cover significantly increased at Sites 1 and 2 after transplanting. In the first year, transplanting significantly increased native forb and grass richness (Table 2). Transplanting introduced 13 native species to Site 1, which had lowest native diversity prior to treatment, 8 species at Site 2, and 5 species at Site 3, which had highest initial native diversity. Keystone species *F. campestris* and *F. idahoensis* were introduced to all sites. In the second year, however, native cover and richness declined at most sites to levels similar to prior to transplanting with some significant changes (Table 2). Key species to the community persisted on sites during the 14-month study period.

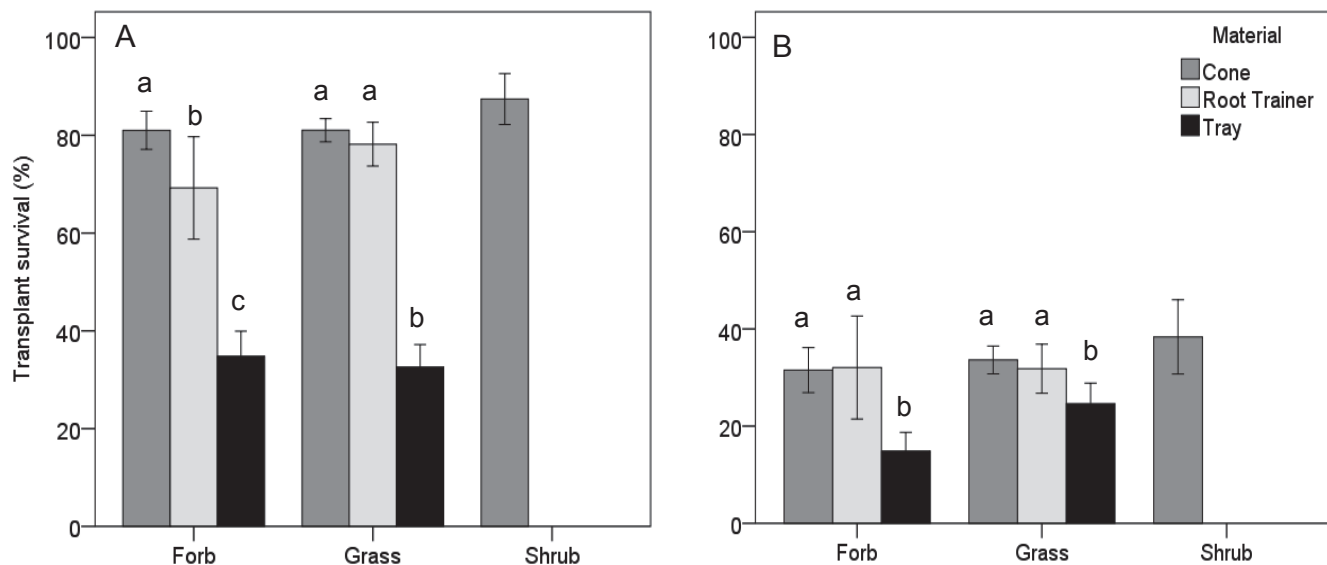


Figure 3. Effect of container size on transplant survival (percent live plants of total planted) at three disturbed sites in (A) July 2012 and (B) July 2013. Within functional groups, columns that do not share a common letter are significantly different. Error bars are 95% confidence intervals.

Discussion

Transplanting was effective at facilitating restoration of fescue grasslands, particularly in introducing species that would not otherwise colonize disturbed sites. Keystone species such as *F. campestris* and *F. idahoensis* are essential for ecological functioning of fescue grasslands, and their presence, even if not yet significant in abundance, can accelerate soil and plant community restoration. Transplant performance was strongly species dependent, as found in previous research (Davies et al. 1999, Page and Bork 2005). Species producing seedlings with durable, well developed root systems such as *K. macrantha* and *Festuca* (fescue) bunch grasses or rhizomatous species such as *H. villosa*, *A. millefolium*, and *A. michauxiana* had greatest survival, similar to results found for these species in other studies (Paschke et al. 2000, Ewing 2002). Mid- to late-successional shrubs performed well as cone transplants, and this may be the preferred method to reintroduce them to restoration sites as they often perform poorly from seed once initial vegetation has established.

Plant container type significantly impacted revegetation success, and larger seedlings with greater root mass are necessary to promote health and survival. Larger size leading to greater transplant survival is consistent with several studies (Davies et al. 1999, Steed and DeWald 2003, Page and Bork 2005, Halsey et al. 2017), although there are exceptions (Bull et al. 2004). In this study, root depths of 10 to 15 cm were necessary for transplant survival. This size may be at the threshold of becoming too large, as rocky soils of foothills fescue grasslands do not allow for deeper holes during planting. The lower winter mortality of grass tray seedlings is likely due to an initial higher seedling density in this container type; even if half died, many remain for the second growing season. This did not, however, benefit the forbs nor did it improve the overall survival of grass tray transplants.

Management to control non-native plants on a restoration site can enhance transplant survival if canopy cover is reduced, as with glyphosate in this study. Glyphosate and cutting are known to increase native transplant survival by reducing resident vegetation (Wilson and Gerry 1995,

Table 2. Mean canopy cover $\pm$ standard error (percent) and mean plot species richness $\pm$ standard error of native species prior to (2011) and in the first two growing seasons (2012, 2013) following transplanting on three disturbed sites. Plot size was 216 m² at Site 1 and 250 m² at Sites 2 and 3. Means within functional groups (forb, grass, shrub) followed by different letters are significantly different ($p < 0.05$).

		Forbs			Grasses			Shrubs		
		2011	2012	2013	2011	2012	2013	2011	2012	2013
Site 1	Cover	0.9 $\pm$ 0.3 ^a	5.2 $\pm$ 0.9 ^b	2.4 $\pm$ 0.8 ^{ac}	0.6 $\pm$ 0.3	1.0 $\pm$ 0.4	0.3 $\pm$ 0.1	0.0 $\pm$ 0.0 ^a	0.1 $\pm$ 0.0 ^b	0.0 $\pm$ 0.0 ^b
	Richness	2.0 $\pm$ 0.0 ^a	3.0 $\pm$ 0.0 ^b	1.0 $\pm$ 0.0 ^c	1.0 $\pm$ 0.0 ^a	2.0 $\pm$ 0.0 ^a	1.0 $\pm$ 0.0 ^b	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Site 2	Cover	2.7 $\pm$ 1.0 ^a	12.9 $\pm$ 2.1 ^b	5.8 $\pm$ 1.3 ^a	0.1 $\pm$ 0.0 ^a	2.0 $\pm$ 0.5 ^b	1.0 $\pm$ 0.5 ^{ab}	0.1 $\pm$ 0.1 ^a	1.1 $\pm$ 0.4 ^b	0.4 $\pm$ 0.2 ^{ab}
	Richness	5.0 $\pm$ 1.0 ^a	10.0 $\pm$ 1.0 ^b	5.0 $\pm$ 1.0 ^a	2.0 $\pm$ 0.0 ^a	4.0 $\pm$ 0.0 ^b	2.0 $\pm$ 0.0 ^a	0.0 $\pm$ 0.0	1.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Site 3	Cover	5.6 $\pm$ 1.3 ^a	14.2 $\pm$ 1.3 ^b	15.1 $\pm$ 1.3 ^b	2.0 $\pm$ 0.1	3.2 $\pm$ 0.7	2.1 $\pm$ 0.4	0.5 $\pm$ 0.2	1.1 $\pm$ 0.7	0.4 $\pm$ 0.3
	Richness	13.0 $\pm$ 1.0 ^a	18.0 $\pm$ 1.0 ^b	13.0 $\pm$ 1.0 ^a	2.0 $\pm$ 0.0 ^a	4.0 $\pm$ 0.0 ^b	3.0 $\pm$ 0.0 ^{ab}	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

Stromberg and Kephart 1996, Huddleston and Young 2005, Jusaitis and Polomka 2008, Middleton et al. 2010, Holl et al. 2014). Cutting may be an environmentally sensitive alternative to herbicide to reduce non-native plant competition, but it negatively affected transplant survival and grass flowering in fescue grassland. Lower overwinter mortality in plots treated with glyphosate may reflect that plants were healthier in this treatment at the end of the first growing season due to reduced competition with non-native plants. Williams et al. (2007) reported release from competition resulted in forb plugs with greater above and below ground growth and therefore survival. Efforts in the first year to improve transplant growth such as addition of mulch could reduce overwinter mortality. This research showed transplant survival will depend on specific conditions at a restoration site. Grasslands with higher biomass and lower species diversity often have lower transplant survival (Davies et al. 1999, Middleton et al. 2010). Mortality increased with time, with high overwinter losses for all species and container types. For long-term success, follow-up non-native plant control is likely critical to reduce seedling mortality by competition with non-native plants, in particular perennial grasses.

Conclusion

Control of non-native plants and ecosystem restoration are imperative for conserving biodiversity and represent two of the greatest challenges for land managers this century (D'Antonio and Meyerson 2002). In this study, transplanting provided a viable method of establishing native species during fescue grassland restoration. Pre-treatment with glyphosate to reduce pre-existing non-native cover increased transplant survival, but cutting and steaming did not. More sustainable alternatives to herbicide that can promote transplant survival should be further researched, and long-term studies are necessary to confirm these short term trends. Container type significantly affected grass and forb transplant survival: survival was higher for larger container stock. A plug with sufficient root size is necessary, but it must not be too large in order to avoid difficulties planting in shallow or rocky soils and to avoid inefficient use of resources. Transplanting should be recognized as an important tool for restoring and maintaining biodiversity in grassland ecosystems.

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